

 B.TECH 1ST YEAR · 1ST SEMESTER · EXAM REVISION

BASIC CIVIL & MECHANICAL ENGINEERING

BCME – Complete Revision Document

UNIT III

Transportation Engineering & Water Resources and Environmental Engineering

 Importance of Transportation

 Highway Engineering

 Types of Pavements

 Harbour Engineering

 Tunnel Engineering

 Airport Engineering

 Railway Engineering

 Water Resources

 Hydrology

 Rainwater Harvesting

 Dams & Reservoirs

 Flashcards & Quick Revision

Contains: Definitions · Tables · Comparisons · Flashcards · Exam Tips · Quick Revision Sheet



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1

 IMPORTANCE OF TRANSPORTATION

Key Roles

- **Economic Development:** Connects producers to markets; boosts GDP
- **Trade & Commerce:** Enables movement of raw materials and finished goods
- **Rural–Urban Connectivity:** Reduces urban–rural divide; provides access to services
- **Defense & Emergency:** Rapid troop deployment; disaster response logistics
- **Industrial Growth:** Movement of machinery, resources, and labour
- **Social & Cultural:** Tourism, migration, cultural exchange

Aspect	Role of Transportation
Economy	Boosts trade, reduces cost
Agriculture	Market access for farmers
Industry	Supply chain & logistics
Defense	Strategic mobility
Social	Employment & education access
Culture	Tourism & integration

2

 TYPES OF TRANSPORTATION SYSTEMS

Mode	Features	Advantages	Limitations
Roadways	Flexible routes, door-to-door, all-terrain	Cheapest for short distances; most accessible	Slow for long distances; weather-dependent
Railways	Fixed track, high capacity, scheduled	Economical for bulk goods; low fuel cost per ton	High infrastructure cost; inflexible routes
Airways	Fastest; international connectivity	Speed; suitable for perishables & passengers	Very expensive; limited cargo capacity
Waterways	Cheapest per ton-km; uses rivers/sea	Bulk cargo; lowest cost; no infrastructure needed	Slowest; route-limited; weather risk

3

 HIGHWAY ENGINEERING BASICS

Definition

A **highway** is a main public road connecting cities, towns, or important places, designed for high-speed motor vehicle traffic.

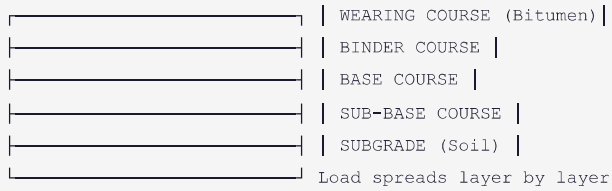
Classification of Roads (India – Nagpur Classification)

- **National Highways (NH):** Connect state capitals, major ports; maintained by NHAI
- **State Highways (SH):** Connect district HQs within a state
- **District Roads:** Connect taluk/tehsil to district HQs
- **Village Roads:** Connect villages to nearby roads/towns

ROAD HIERARCHY (India) ————— NATIONAL
HIGHWAYS (NH) | STATE HIGHWAYS (SH) | DISTRICT ROADS |
VILLAGE ROADS ————— NH → Highest
priority/standard VR → Lowest standard/cost

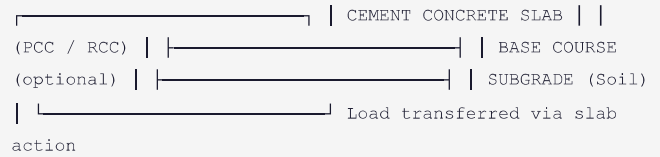


Flexible Pavement



- **Material:** Bituminous/asphalt concrete
- **Load:** Distributes to lower layers gradually
- **Flexes** under load without cracking
- **Life:** 10–15 years

Rigid Pavement



- **Material:** Portland cement concrete
- **Load:** Distributed over wide area via slab
- **Does not flex;** resists deformation
- **Life:** 25–40 years

⚡ Difference Table: Flexible vs Rigid Pavement

MANDATORY

Feature	Flexible Pavement	Rigid Pavement
Material	Bitumen / Asphalt	Cement Concrete (PCC/RCC)
Load Distribution	Layer by layer (flexible)	Slab action (wide spread)
Design Life	10–15 years	25–40 years
Initial Cost	Low	High
Maintenance Cost	High (frequent repairs)	Low
Surface Texture	Smooth, dark/black	Hard, grey/white
Temperature Effect	Softens in heat	Expands (needs joints)
Traffic Suitability	Light to medium	Heavy traffic
Repair	Easy, patchwork possible	Difficult & costly
Examples	City roads, NH (older)	Highways, runways, bridges



Harbour: A sheltered body of water where ships, boats, and vessels can anchor safely.

Types

- **Natural Harbour:** Naturally sheltered by landforms (e.g., Mumbai, Visakhapatnam)
- **Artificial Harbour:** Man-made with breakwaters and structures (e.g., Ennore Port)

Components

Component	Function
Breakwater	Protects harbour from waves/storms
Dock	Enclosed area for loading/unloading ships

HARBOUR LAYOUT (Text Diagram)



Pier	Platform extending into water for ship berthing
Wharf	Structure along shore for loading operations
Jetty	Narrow structure into sea for mooring
Quay	Solid-built bank for mooring alongside



Tunnel: An underground or underwater passageway, typically enclosed, used for transportation or conveyance.

Uses

- Road/railway transportation through mountains
- Water conveyance (irrigation, drainage)
- Sewer and utility tunnels
- Metro rail systems

Types

- **Traffic Tunnels:** Road tunnels, railway tunnels, pedestrian tunnels
- **Hydraulic Tunnels:** Water supply, irrigation, power generation
- **Mining Tunnels:** Access to mineral deposits

Advantages

- Reduces travel distance/time through mountains
- Avoids steep gradients on surface
- Unaffected by weather conditions
- Conserves surface land area
- Reduces noise & visual pollution

Construction Methods (Brief)

- **Cut & Cover:** Trench dug, tunnel built, covered
- **Drill & Blast:** Rock drilled, explosives used
- **TBM (Tunnel Boring Machine):** Machine bores through earth
- **Shield Method:** Used in soft/waterlogged soils



Airport: An aerodrome with facilities for aircraft to take off, land, and be maintained, including terminal buildings for passengers.

Key Components

Component	Function
Runway	Paved strip for aircraft takeoff & landing
Taxiway	Path connecting runway to terminal/hangar
Terminal Building	Passenger processing, check-in, customs
Apron/Tarmac	Aircraft parking area near terminal
Control Tower	Air traffic control (ATC)
Hangar	Aircraft maintenance & storage

Site Selection Factors

- **Topography:** Flat land; no obstacles near approach paths
- **Wind:** Runway aligned with prevailing winds
- **Fog & Weather:** Low fog-prone areas preferred
- **Soil:** Good bearing capacity for pavements
- **Accessibility:** Good road/rail connection to city
- **Noise:** Away from residential areas
- **Airspace:** Free from tall structures, hills, overhead cables
- **Drainage:** Proper surface water drainage



Railway: A guided transport system using rails (tracks) on which trains run, used for mass transportation of passengers and freight.

Key Components

Component	Description
Rails	Steel tracks on which wheels run
Sleepers (Ties)	Transverse supports holding rails; wood/concrete/steel
Ballast	Crushed stone beneath sleepers; provides drainage & stability
Fish Plates	Plates joining two rail ends
Spikes/Keys	Fasten rails to sleepers

Types of Railway Gauges (India)

Gauge	Width	Use
Broad Gauge (BG)	1.676 m	Main lines; most common in India
Metre Gauge (MG)	1.000 m	Older branch lines (being converted)
Narrow Gauge (NG)	0.762 m	Hilly/forest areas; toy trains

Remember

India is converting all routes to **Broad Gauge (1676 mm)** under Project Unigauge for uniformity.



Water Resources Engineering: The branch of civil engineering dealing with collection, storage, control, treatment, distribution, and use of water.

Importance of Water Resources

- Essential for drinking and domestic use
- Agriculture and irrigation (70% of use)
- Industrial processes and cooling
- Hydropower generation
- Navigation and recreation
- Ecosystem and biodiversity support

Need for Management

- Uneven distribution of rainfall across regions
- Growing population → increasing demand
- Pollution of existing water sources
- Climate change altering rainfall patterns
- Prevent floods and droughts
- Sustainable use for future generations



Surface Water

- **Rivers:** Perennial/seasonal; major irrigation source
- **Lakes:** Natural/artificial; regulated supply
- **Ponds:** Small water bodies; local use
- **Reservoirs:** Man-made; controlled storage

Groundwater

- **Wells:** Dug wells, tube wells, bore wells
- **Springs:** Natural groundwater surface outcrop
- **Artesian Wells:** Water rises under pressure

Source	Examples	Advantages	Limitations
Rivers	Ganga, Krishna	Large volume; perennial	Flood risk; pollution
Lakes	Dal Lake	Regulated; clean	Seasonal variation
Groundwater	Bore wells	Usually clean; reliable	Over-extraction; slow recharge
Rainwater	Collected roof water	Free; pure	Seasonal; limited quantity
Springs	Hill springs	Pure; gravity-fed	Low quantity; location-limited



Physical Characteristics

Property	Standard
Color	Colorless (≤ 5 Hazen units)
Taste	Tasteless
Odour	Odourless
Turbidity	≤ 5 NTU (drinking)
Temperature	10–15°C (ideal)

Chemical Characteristics

Property	Standard
pH	6.5–8.5 (neutral)
Hardness	≤ 300 mg/L
TDS	≤ 500 mg/L
Chlorides	≤ 250 mg/L
Fluorides	0.6–1.2 mg/L

Biological Characteristics

- **Bacteria:** E. coli should be absent
- **Viruses:** Completely absent
- **Algae:** Absent in treated water
- **Protozoa:** Absent (Giardia, Cryptosporidium)

MPN (Most Probable Number): Test to detect coliform bacteria. Should be **0** per 100 mL for drinking water.



Hydrology: The science that deals with the occurrence, distribution, movement, and properties of water on Earth and its atmosphere.

Key Components of Hydrological Cycle

- **Precipitation (P):** Rain, snow, hail falling on Earth
- **Interception:** Water caught by vegetation before reaching ground
- **Infiltration:** Water entering soil from surface
- **Evaporation:** Water from surface changing to vapour
- **Transpiration:** Water vapour released by plants
- **Runoff:** Water flowing over land surface to streams/ivers
- **Groundwater Flow:** Slow subsurface movement of water
- **Condensation:** Water vapour becoming liquid (clouds form)

HYDROLOGICAL CYCLE

[CLOUDS] ← CONDENSATION | PRECIPITATION (Rain/Snow/Hail)
 ↓ | INTERCEPTION (Plants) | |
 INFILTRATION (Soil) | | SURFACE RUNOFF → RIVERS
 ↓ ↓ GROUNDWATER [OCEAN/LAKE] ↑
 ↑ SPRINGS → ↑ EVAPORATION & TRANSPIRATION
 (Back to atmosphere) Water
 Balance: $P = R + E + \Delta S$ P=Precipitation, R=Runoff,
 E=Evapotranspiration, ΔS =Storage change



Rainwater Harvesting: The collection, storage, and use of rainwater for domestic, agricultural, or groundwater recharge purposes.

Methods

- **Rooftop Harvesting:** Rainwater from rooftops collected in tanks/pits
- **Surface Runoff Harvesting:** Runoff collected in check dams, tanks, ponds
- **Recharge Pits:** Dug pits that allow water to percolate to groundwater
- **Percolation Tanks:** Shallow reservoirs for groundwater recharge

Advantages

- Recharges groundwater table
- Reduces dependence on external supply
- Reduces runoff and urban flooding
- Free water source; low cost
- Improves water quality in aquifers
- Useful in water-scarce regions

ROOFTOP RAINWATER HARVESTING

[RAIN] ↓ [ROOFTOP / TERRACE] ↓ [GUTTERS & PIPES] ↓
 [FILTER / MESH] ↓ | FIRST FLUSH | ←
 First rain discarded | DIVERTER | ↓
 [STORAGE TANK] → DOMESTIC USE ↓ [OVERFLOW TO RECHARGE
 PIT] ↓ [GROUNDWATER AQUIFER]



Dams

A **dam** is a barrier constructed across a river or stream to store water, control flow, and harness water resources.

Type of Dam	Material/Feature
Gravity Dam	Concrete; resists water by own weight
Arch Dam	Curved; transfers force to valley walls
Earth Dam	Compacted earth/rock fill
Buttress Dam	Slabs supported by buttresses

Purposes of Dams

- Water storage for irrigation, drinking
- Hydropower generation

Reservoirs

A **reservoir** is an artificial or natural lake where water is stored behind a dam or in a depression, for controlled supply.

Functions

- Store water during high rainfall for dry season
- Regulate river flow
- Supply drinking water to cities
- Support irrigation systems
- Recreation and fisheries

DAM & RESERVOIR STRUCTURE [RIVER] →
 [RESERVOIR] ↑ [DAM WALL] ↓ [SPILLWAY] [SLUICE GATE] ↓ [CANAL /
 RIVER]

- Flood control
- Navigation improvement



15A. Transportation Modes Comparison

Mode	Speed	Cost	Capacity	Best Use
Roadway	Medium (60–100 km/h)	Low–Medium	Medium	Short–medium distances; door-to-door
Railway	Medium-High (100–200 km/h)	Low (bulk)	Very High	Long distance; bulk cargo; passengers
Airway	Very High (800+ km/h)	Very High	Low	Long distance; urgent; perishables
Waterway	Very Low (15–30 km/h)	Very Low	Extremely High	Bulk heavy cargo; international trade

15B. Pavements Comparison

Feature	Flexible Pavement	Rigid Pavement
Binder	Bitumen/Asphalt	Cement
Load Transfer	Layer by layer (friction)	Beam/slab action
Life	10–15 years	25–40 years
Cost (initial)	Low	High
Maintenance	Frequent, easy	Rare, difficult
Surface	Black/dark, smooth	Grey/white, rough
Temperature effect	Softens in heat	Expands; needs joints
Where used	City roads, flexible routes	Highways, runways, bridges

15C. Water Sources Comparison

Source	Availability	Quality	Treatment Need
River	High (perennial)	Variable; often polluted	Full treatment needed
Lake	Moderate	Better than rivers	Moderate treatment
Groundwater	Reliable; season-independent	Usually good; may be hard	Softening & disinfection
Rainwater	Seasonal only	Purest form	Minimal (filter + disinfect)
Spring	Low to moderate	Good	Minimal

15D. Engineering Fields Comparison

Field	Purpose	Key Structure / Example
Highway Engineering	Road design & construction	NH, SH, Expressways
Railway Engineering	Rail track design	Broad gauge lines, Metro
Airport Engineering	Aviation infrastructure	Runway, Terminal, ATC
Harbour Engineering	Port & marine structures	Breakwater, Dock, Pier
Tunnel Engineering	Underground passageways	Road/rail/hydraulic tunnels
Water Resources Engg.	Water collection, storage, use	Dam, Reservoir, Canal



Highway → Main public road connecting towns/cities for high-speed traffic

Flexible Pavement → Bituminous road that distributes load layer by layer

Harbour → Sheltered body of water where ships can dock safely

Dock → Enclosed water area for ship loading/unloading

Runway → Paved strip at airport for aircraft takeoff and landing

Gauge → Distance between inner faces of two parallel rails

Ballast → Crushed stone under sleepers providing stability and drainage

Hydrological Cycle → Continuous movement of water through evaporation, condensation, precipitation, and runoff

Runoff → Water flowing over land surface after rainfall, reaching streams

Evapotranspiration → Total water loss from surface via evaporation + plant transpiration

Dam → Barrier across a river to store water and control flow

Spillway → Structure in a dam to release excess water safely

Turbidity → Cloudiness/haziness of water due to suspended particles

Hardness → Presence of dissolved salts of Ca and Mg in water

National Highway → Road maintained by NHAI connecting state capitals and major ports

Wharf → Structure built along shore for ship loading/unloading

Percolation → Movement of water through soil/rock layers downward to groundwater

Pavement → Hard surface layer of a road designed to bear and distribute traffic loads

Rigid Pavement → Cement concrete road that distributes load via slab action

Breakwater → Offshore structure protecting harbour from wave action

Tunnel → Underground passageway for transportation or water conveyance

Taxiway → Path connecting runway to terminal or hangar

Sleeper → Transverse support member on which rails are fastened

Hydrology → Science of water occurrence, distribution, and movement on Earth

Precipitation → Any form of water (rain, snow, hail) falling from atmosphere

Infiltration → Downward movement of water from soil surface into the ground

Aquifer → Underground rock layer that holds and transmits groundwater

Reservoir → Artificial lake created behind a dam for water storage

Rainwater Harvesting → Collection and storage of rainwater for future use or groundwater recharge

pH → Measure of acidity/alkalinity of water (6.5–8.5 for drinking)

TDS → Total Dissolved Solids in water; ≤500 mg/L for drinking

Pier → Platform extending into water for ship berthing

Broad Gauge → Railway track width of 1.676 m; most common in India

Apron/Tarmac → Paved area at airport where aircraft park near terminal



💧 Water Balance Equation (Hydrology)

$$P = R + ET + \Delta S$$

P = Precipitation (mm)

R = Runoff (mm)

ET = Evapotranspiration (mm)

ΔS = Change in Storage

🌧️ Rational Formula (Runoff Estimation)

$$Q = C \times I \times A / 360$$

Q = Peak runoff (m^3/s)

C = Runoff coefficient (0-1)

I = Rainfall intensity (mm/hr)

A = Catchment area (hectares)

360 = Unit conversion constant

🏠 Rooftop Rainwater Volume

$$V = A \times R \times C$$

V = Volume of water collected (litres)

A = Roof area (m^2)

R = Annual rainfall (mm)

C = Collection coefficient (0.7-0.9)

👤 Per Capita Water Demand

$$D = P \times d$$

D = Total demand (litres/day)

P = Population

d = Per capita demand (135 L/day for urban India)

📏 Road Grade / Gradient

$$\text{Grade (\%)} = (\text{Rise} / \text{Run}) \times 100$$

Rise = Vertical height gained (m)

Run = Horizontal distance (m)

Max gradient on NH = 3.3%

🌊 Catchment Water Yield

$$\text{Yield} = P \times A \times C$$

P = Annual precipitation (m)

A = Catchment area (m^2)

C = Runoff coefficient

★ Key Constants to Remember

- Broad Gauge width = **1.676 m**
- Drinking water pH = **6.5–8.5**
- Per capita demand (urban) = **135 L/day**
- Max turbidity = **5 NTU** (drinking)
- Max TDS = **500 mg/L** (drinking)
- Max hardness = **300 mg/L** (drinking)



Q What is flexible pavement?

A Bituminous pavement that distributes load gradually layer by layer to subgrade

Q What is hydrology?

A Science of water movement, distribution, and occurrence on Earth

Q What is a harbour?

A Sheltered coastal water area where ships can dock and anchor safely

Q What is rainwater harvesting?

A Collection and storage of rainwater for use or groundwater recharge

Q What is a tunnel?

A Underground/underwater passageway for transportation or water conveyance

Q What is a dam?

A Barrier across a river to store water, control floods, and generate power

Q What is a National Highway?

A Road connecting state capitals/major ports; maintained by NHAI

Q What is turbidity?

A Cloudiness of water due to suspended particles; max 5 NTU for drinking

Q What is the ideal pH for drinking water?

A 6.5 to 8.5 (neutral to slightly alkaline)

Q What is infiltration?

A Downward movement of water from soil surface to subsurface/groundwater

Q What is per capita water demand in India (urban)?

A 135 litres per person per day (CPHEEO standard)

Q What is a wharf?

A Structure built along shore for loading/unloading ships

Q What is TBM in tunnel construction?

A Tunnel Boring Machine – machine that drills tunnels mechanically

Q What is an aquifer?

A Underground permeable rock/sediment layer storing and transmitting groundwater

Q What is the Rational Formula for runoff?

A $Q = CIA/360$ (Q =peak runoff, C =coefficient, I =intensity, A =area)

Q What is rigid pavement?

A Cement concrete pavement that distributes load via slab/beam action

Q What is runoff?

A Water that flows over land surface after rainfall, reaching rivers/streams

Q What is a breakwater?

A Offshore structure protecting harbour from destructive wave action

Q What is Broad Gauge?

A Railway track gauge of 1.676 m; most common gauge in India

Q What is the hydrological cycle?

A Continuous cycle: Precipitation → Runoff → Evaporation → Condensation → Precipitation

Q What is a reservoir?

A Artificial lake formed behind a dam for controlled water storage

Q What is ballast in railway?

A Crushed stone beneath sleepers providing drainage, stability, and load distribution

Q What is a runway?

A Paved strip at airport for aircraft takeoff and landing operations

Q What is a gravity dam?

A Concrete dam that resists water pressure by its own weight alone

Q Name layers of flexible pavement (top to bottom)

A Wearing course → Binder course → Base course → Sub-base → Subgrade

Q What is water hardness?

A Presence of dissolved calcium and magnesium salts; max 300 mg/L for drinking

Q Which mode of transport is cheapest per ton-km?

A Waterways (sea/river transport)

Q What is evapotranspiration?

A Combined water loss via evaporation from soil + transpiration from plants

Q What is a sleeper in railways?

A Transverse support member (wood/concrete/steel) holding rails in position

Q What is a taxiway?

A Paved path connecting the runway to terminal, hangar, or apron

**✗ Common Mistakes to Avoid****1. Confusing Flexible & Rigid Pavement**

Remember: Flexible = Black + Bitumen + Layer-by-layer. Rigid = Grey + Cement + Slab action.

2. Missing Components in Questions

When asked about Airport/Harbour/Railway — always list ALL components (not just 2–3).

3. Writing Long Theory

Use bullet points; examiners reward conciseness + structure over paragraphs.

4. Forgetting Units in Formulas

Always write units: Q in m^3/s , I in mm/hr , A in hectares.

5. Missing Diagrams

Even a simple labeled box diagram earns marks. Draw pavement layers, harbour layout, hydrological cycle.

6. Confusing Pier, Wharf, Quay

Pier = extends INTO water. Wharf = ALONG shore. Quay = solid-built bank.

✓ Writing Tips for High Marks**1. Use Tables for Comparison Questions**

Draw a 2-column comparison table. Never write "Flexible pavement is... whereas rigid..."

2. Draw Simple Diagrams

Pavement layers, harbour layout, hydrological cycle — even rough diagrams earn marks.

3. Highlight Keywords in Definitions

Start definitions with the key term. "A dam is a barrier..." not "It is used for..."

4. Structure Every Answer

Definition → Types → Components → Advantages → Diagram (for 5-mark questions)

5. Answer Comparison Qs in Tables

10–12 row comparison tables with proper headings = full marks guaranteed.

6. Remember Standard Values

pH 6.5–8.5, TDS ≤ 500 , Turbidity ≤ 5 NTU, BG = 1.676 m — these fetch easy marks.

🕒 Time Management:

- 2-mark: Definition only (2–3 lines)
- 5-mark: Definition + Types + 1 diagram
- 10-mark: Full structure + tables + diagrams

**🛣️ HIGHWAY TYPES**

- NH – National Highways
- SH – State Highways
- DR – District Roads
- VR – Village Roads

🏗️ PAVEMENT LAYERS (FLEX)

- Wearing Course
- Binder Course
- Base Course
- Sub-base Course
- Subgrade

⚓ HARBOUR COMPONENTS

- Breakwater
- Dock
- Pier
- Wharf
- Jetty, Quay

✈️ AIRPORT COMPONENTS

- Runway
- Taxiway
- Terminal Building
- Apron / Tarmac
- Control Tower
- Hangar

🚂 RAILWAY GAUGES

- BG = 1.676 m
- MG = 1.000 m
- NG = 0.762 m

🚂 RAILWAY COMPONENTS

- Rails
- Sleepers
- Ballast
- Fish Plates
- Spikes/Keys

💧 WATER SOURCES

- Rivers, Lakes
- Ponds, Reservoirs
- Groundwater
- Wells, Springs
- Rainwater

🧪 WATER QUALITY

- pH: 6.5–8.5
- TDS: ≤ 500 mg/L
- Turbidity: ≤ 5 NTU
- Hardness: ≤ 300 mg/L
- E. coli: Absent

🌧️ HYDROLOGICAL CYCLE

- Precipitation
- Interception
- Infiltration
- Runoff
- Evapotranspiration
- Condensation

🌱 RWH METHODS

- Rooftop Harvesting
- Surface Runoff
- Recharge Pits
- Percolation Tanks
- Check Dams

🏰 DAM TYPES

- Gravity Dam
- Arch Dam
- Earth Dam
- Buttress Dam

📐 KEY FORMULAS

- $P = R + ET + \Delta S$
- $Q = CIA / 360$
- $V = A \times R \times C$
- $D = P \times d$

🔑 KEY DIFFERENCES (One-Line)**Flexible vs Rigid**

Material	Bitumen vs Cement Concrete
Load	Layer-by-layer vs Slab action
Life	10–15 yrs vs 25–40 yrs
Cost	Low initial vs High initial

Transport Modes Speed Rank

Fastest	Airways (800+ km/h)
Fast	Railways (100–200 km/h)
Medium	Roadways (60–100 km/h)
Slowest	Waterways (15–30 km/h)

🏆 ALL THE BEST FOR YOUR EXAM! 🏆

Remember: Structure + Tables + Diagrams + Key Terms = Top Marks